

Exercise 1: Configuring a Basic Spring Application

```
<dependencies>
<dependency>
  <groupId>junit</groupId>
  <artifactId>junit</artifactId>
  <version>3.8.1</version>
  <scope>test</scope>
</dependency>
<dependency>
  <groupId>org.springframework</groupId>
  <artifactId>spring-context</artifactId>
  <version>5.3.39</version>
</dependency>
</dependencies>
```

BookRepository.java

```
package org.library.repository;

public class BookRepository {
    public void saveBook(String title) {
        System.out.println("Repository Action: Book " + title + " has been archived.");
    }
}
```

BookService.java

```
package org.library.service;

import org.library.repository.BookRepository;

public class BookService {
    private BookRepository bookRepository;
```

```

        public void setBookRepository(BookRepository bookRepository) {
this.bookRepository = bookRepository;
        }

        public void addBook(String title) {
System.out.println("BookService Log: Preparing to archive new book.");
bookRepository.saveBook(title);
        }
}

```

Main.java

```

package org.library;

import org.library.service.BookService;
import org.springframework.context.ApplicationContext;
import org.springframework.context.support.ClassPathXmlApplicationContext;

public class Main {
    public static void main(String[] args) {
        ApplicationContext context = new
ClassPathXmlApplicationContext("applicationContext.xml");

        BookService bookService = (BookService) context.getBean("bookService");
        bookService.addBook("Spring in Action");
    }
}

```

resources/applicationContext.xml

```

<?xml version="1.0" encoding="UTF-8"?>
<beans xmlns="http://www.springframework.org/schema/beans"

```

```
xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
xsi:schemaLocation="
    http://www.springframework.org/schema/beans
    http://www.springframework.org/schema/beans/spring-beans.xsd">
```

```
<bean id="bookRepository" class="org.library.repository.BookRepository"/>
<bean id="bookService" class="org.library.service.BookService">
    <property name="bookRepository" ref="bookRepository"/>
</bean>
```

</beans>

OUTPUT:



```
Run Main x
"C:\Program Files\Eclipse Adoptium\jdk-17.0.12.7-hotspot\bin\java.exe" "-javaagent:C:\Pro
BookService Log: Preparing to archive new book.
Repository Action: Book 'Spring in Action' has been archived.
Process finished with exit code 0
```

Exercise 2: Implementing Dependency Injection

resources/applicationContext.xml

```
<?xml version="1.0" encoding="UTF-8"?>
```

```
<beans xmlns="http://www.springframework.org/schema/beans"
       xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
       xsi:schemaLocation="
           http://www.springframework.org/schema/beans
           http://www.springframework.org/schema/beans/spring-beans.xsd">
    <bean id="bookRepository" class="org.library.repository.BookRepository"/>
    <bean id="bookService" class="org.library.service.BookService">
        <property name="bookRepository" ref="bookRepository"/>
    </bean>
</beans>
```

BookRepository.java

```
package org.library.repository;

public class BookRepository {
    public void saveBook(String title) {
        System.out.println("Repository Info: Book " + title + " successfully saved in the
        database.");
    }
}
```

BookService.java

```
package org.library.service;

import org.library.repository.BookRepository;

public class BookService {
    private BookRepository bookRepository;
```

```

        public void setBookRepository(BookRepository bookRepository) {
this.bookRepository = bookRepository;
        }

        public void addBook(String title) {
System.out.println("Service Alert: Initiating save process for the new book entry.");
bookRepository.saveBook(title);
        }
}

```

Main.java

```

package org.library;

import org.library.service.BookService;
import org.springframework.context.ApplicationContext;
import org.springframework.context.support.ClassPathXmlApplicationContext;

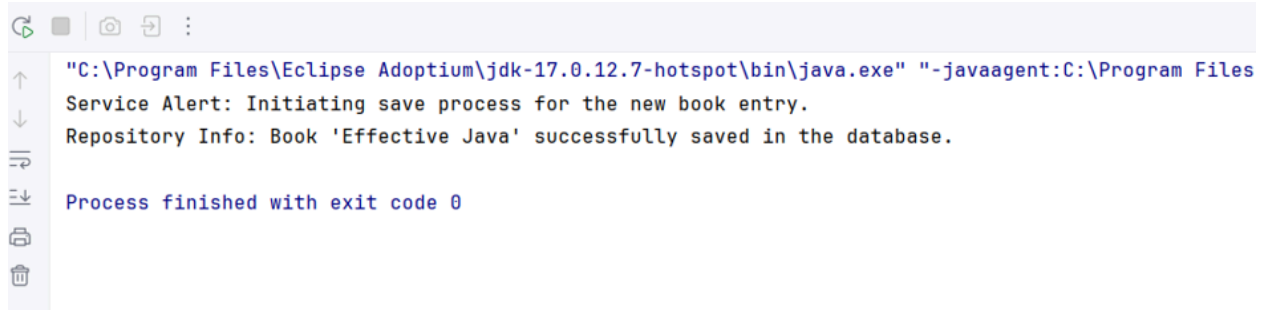
public class Main {
    public static void main(String[] args) {
        ApplicationContext context = new
ClassPathXmlApplicationContext("applicationContext.xml");

        BookService bookService = (BookService) context.getBean("bookService");

        bookService.addBook("Effective Java");
    }
}

```

Output :



```
"C:\Program Files\Eclipse Adoptium\jdk-17.0.12.7-hotspot\bin\java.exe" "-javaagent:C:\Program Files  
Service Alert: Initiating save process for the new book entry.  
Repository Info: Book 'Effective Java' successfully saved in the database.  
  
Process finished with exit code 0
```

Exercise 4: Creating and Configuring a Maven Project

Exercise 4: Creating and Configuring a Maven Project

pom.xml

```
<project xmlns="http://maven.apache.org/POM/4.0.0"  
    xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"  
    xsi:schemaLocation="http://maven.apache.org/POM/4.0.0  
        http://maven.apache.org/xsd/maven-4.0.0.xsd">  
    <modelVersion>4.0.0</modelVersion>  
  
    <groupId>com.library</groupId>  
    <artifactId>LibraryManagement</artifactId>  
    <version>1.0-SNAPSHOT</version>  
  
    <properties>  
        <java.version>1.8</java.version>  
        <spring.version>5.3.34</spring.version>  
        <servlet.version>4.0.1</servlet.version>  
    </properties>
```

```
<dependencies>
  <dependency>
    <groupId>org.springframework</groupId>
    <artifactId>spring-context</artifactId>
    <version>${spring.version}</version>
  </dependency>
  <dependency>
    <groupId>org.springframework</groupId>
    <artifactId>spring-aop</artifactId>
    <version>${spring.version}</version>
  </dependency>
  <dependency>
    <groupId>org.springframework</groupId>
    <artifactId>spring-webmvc</artifactId>
    <version>${spring.version}</version>
  </dependency>
  <dependency>
    <groupId>javax.servlet</groupId>
    <artifactId>javax.servlet-api</artifactId>
    <version>${servlet.version}</version>
    <scope>provided</scope>
  </dependency>
</dependencies>

  <build>
<pluginManagement>
  <plugins>
    <plugin>
      <groupId>org.apache.maven.plugins</groupId>
```

```
<artifactId>maven-compiler-plugin</artifactId>
<version>3.10.1</version>
<configuration>
  <source>${java.version}</source>
  <target>${java.version}</target>
</configuration>
</plugin>
</plugins>
</pluginManagement>
</build>
</project>
```